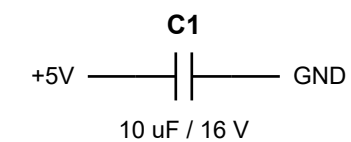
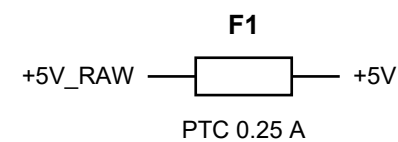
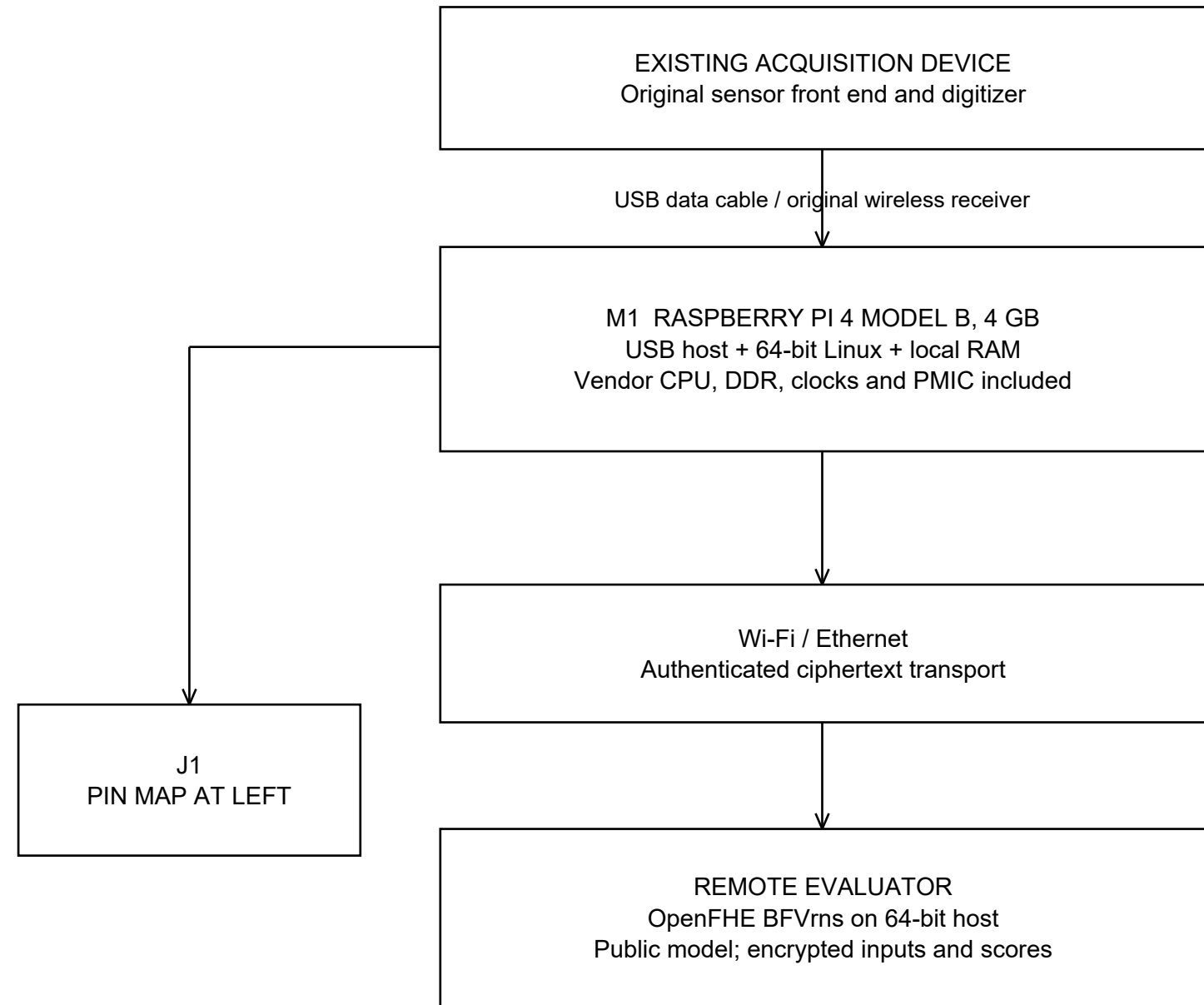
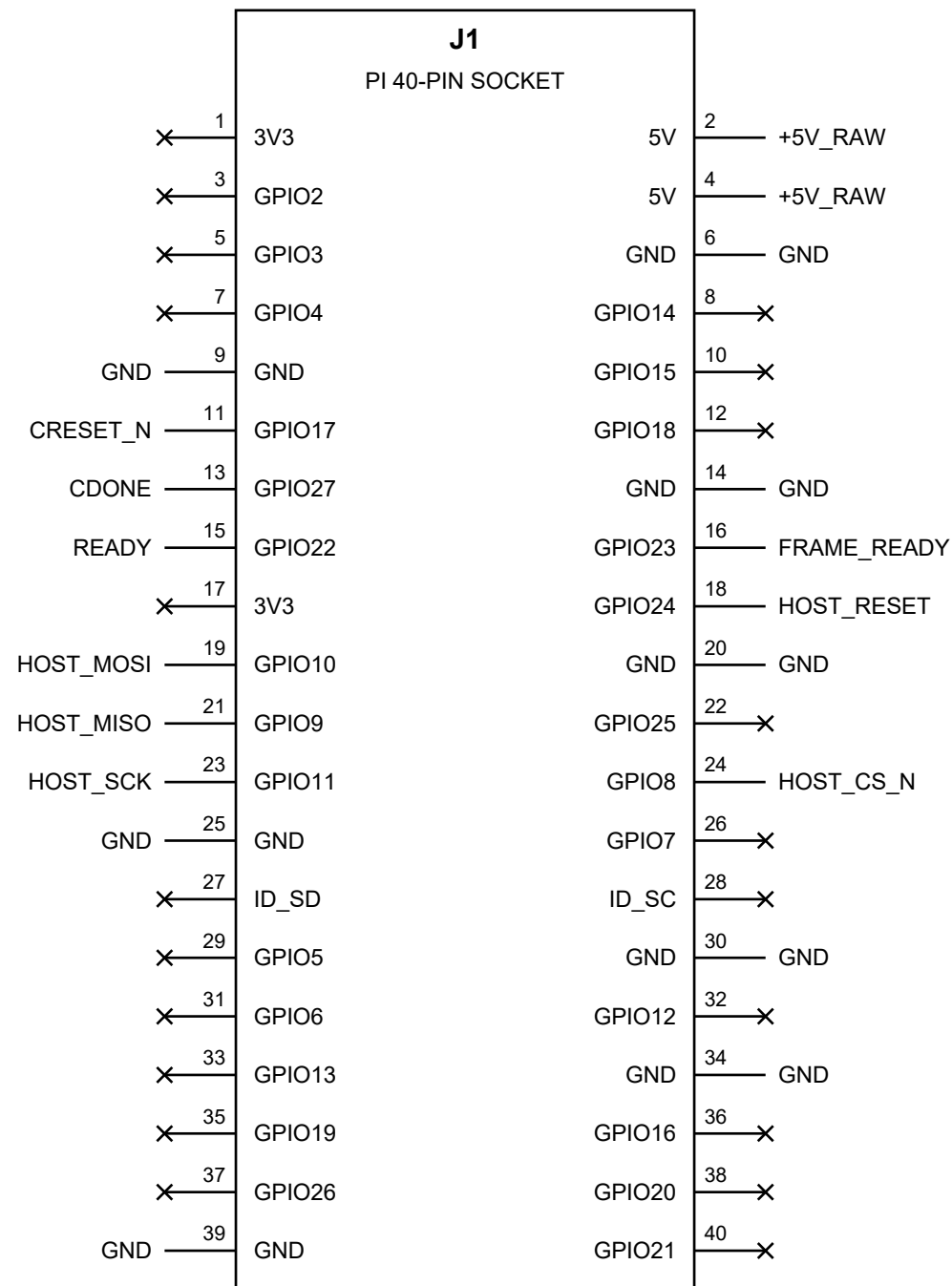
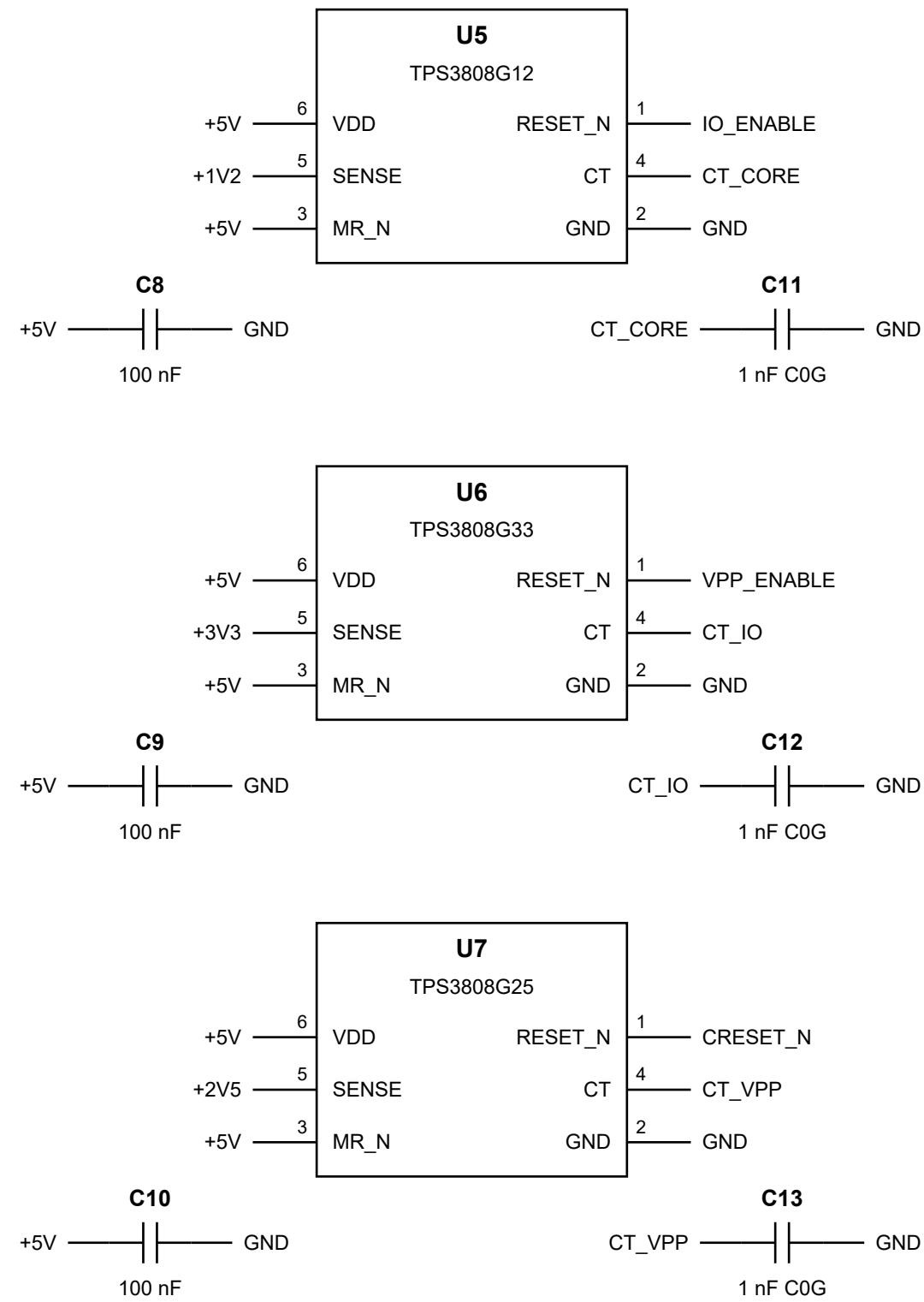
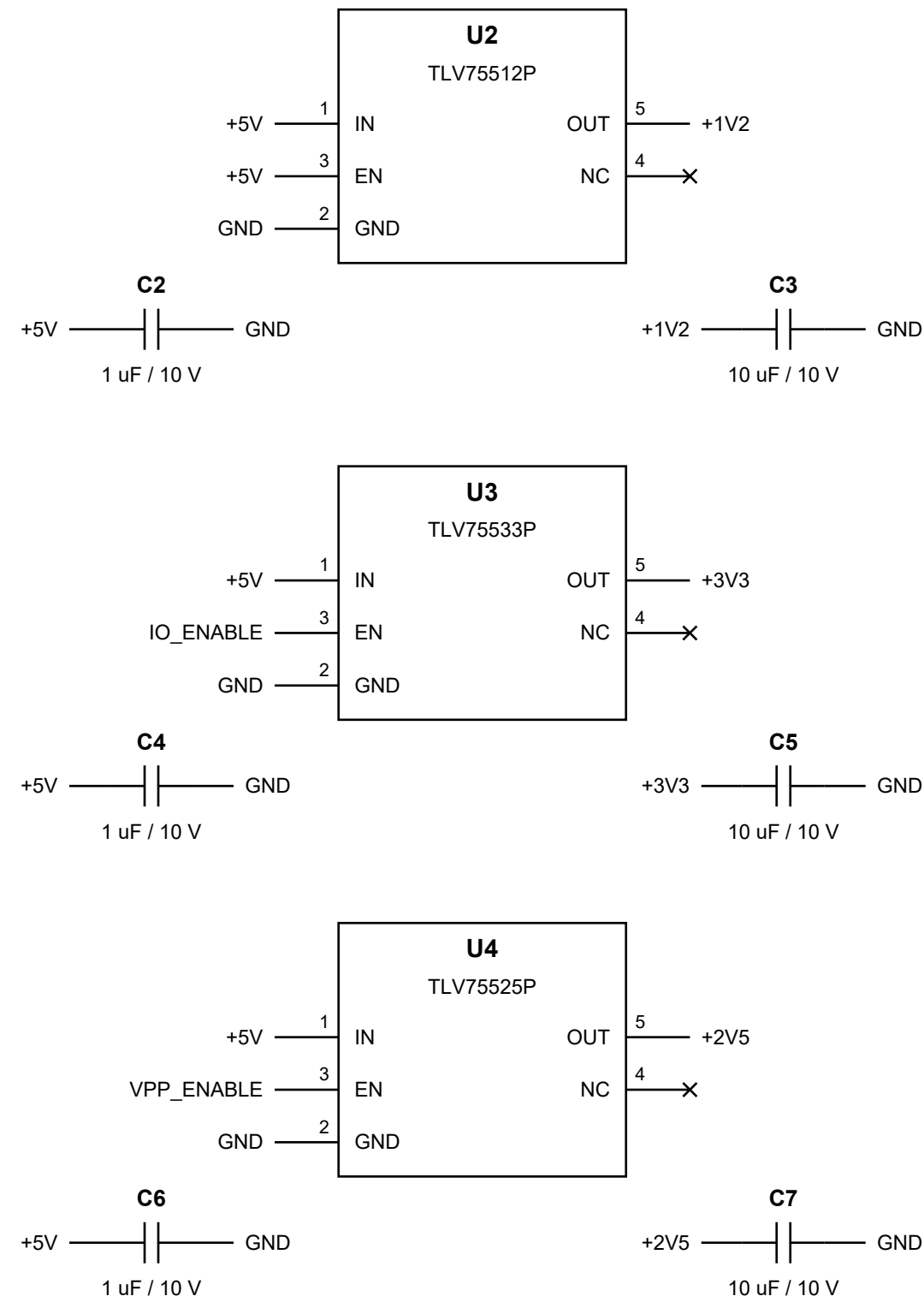


ENER / HOST CONNECTOR AND COMPLETE ASSEMBLY INTERCONNECT



J1 pin numbering is physical, viewed from the mating side; mark pin 1 on the PCB.
M1 is powered through its original USB-C input. Relay draws from J1 pins 2 and 4.
All J1 grounds connect to a continuous relay ground plane. No analog electrode wiring on this board.
3.3 V pins 1 and 17 and all unused GPIO pins are intentionally open. No HAT EEPROM is fitted.

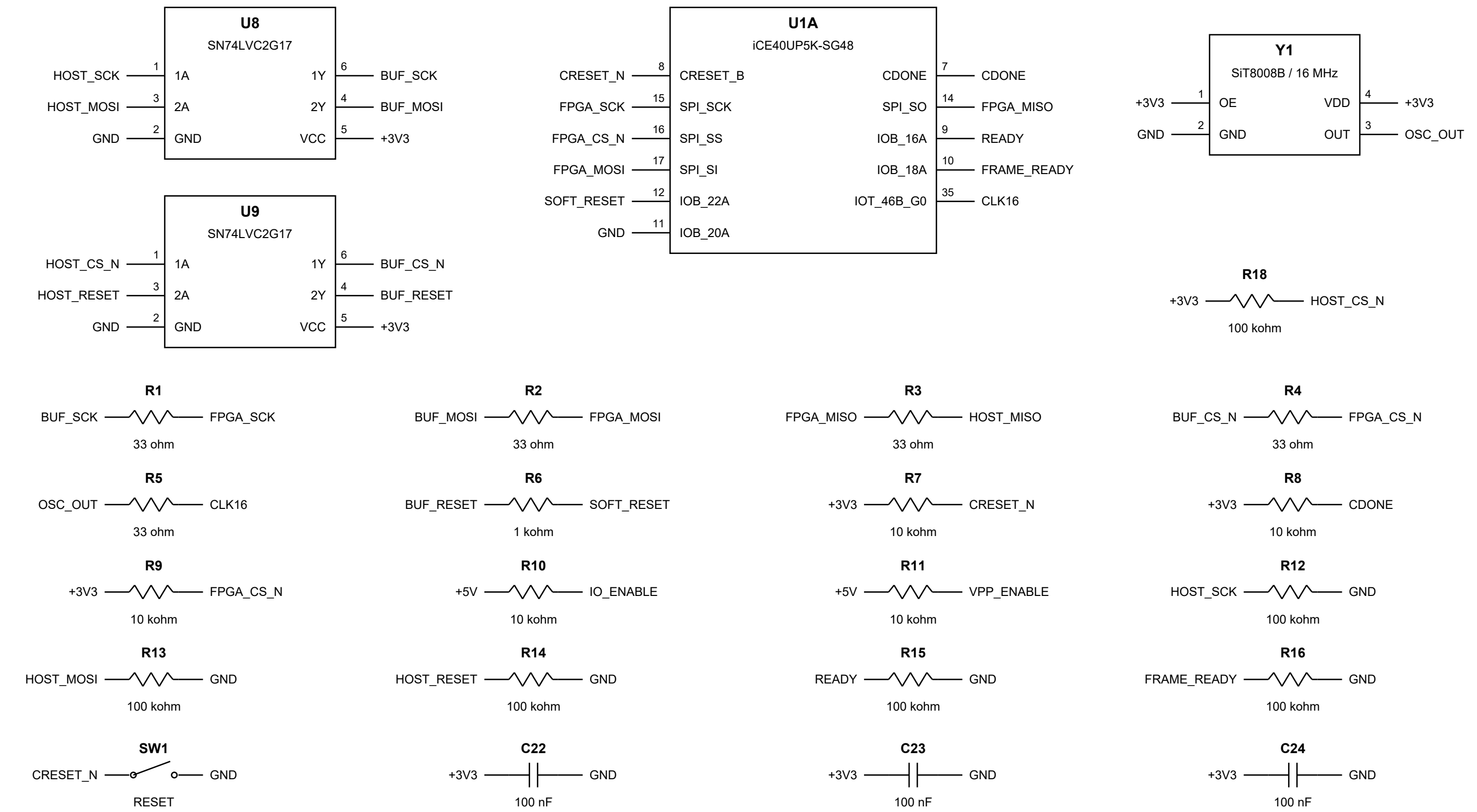
ENER / SEQUENCED POWER RAILS AND RESET SUPERVISION



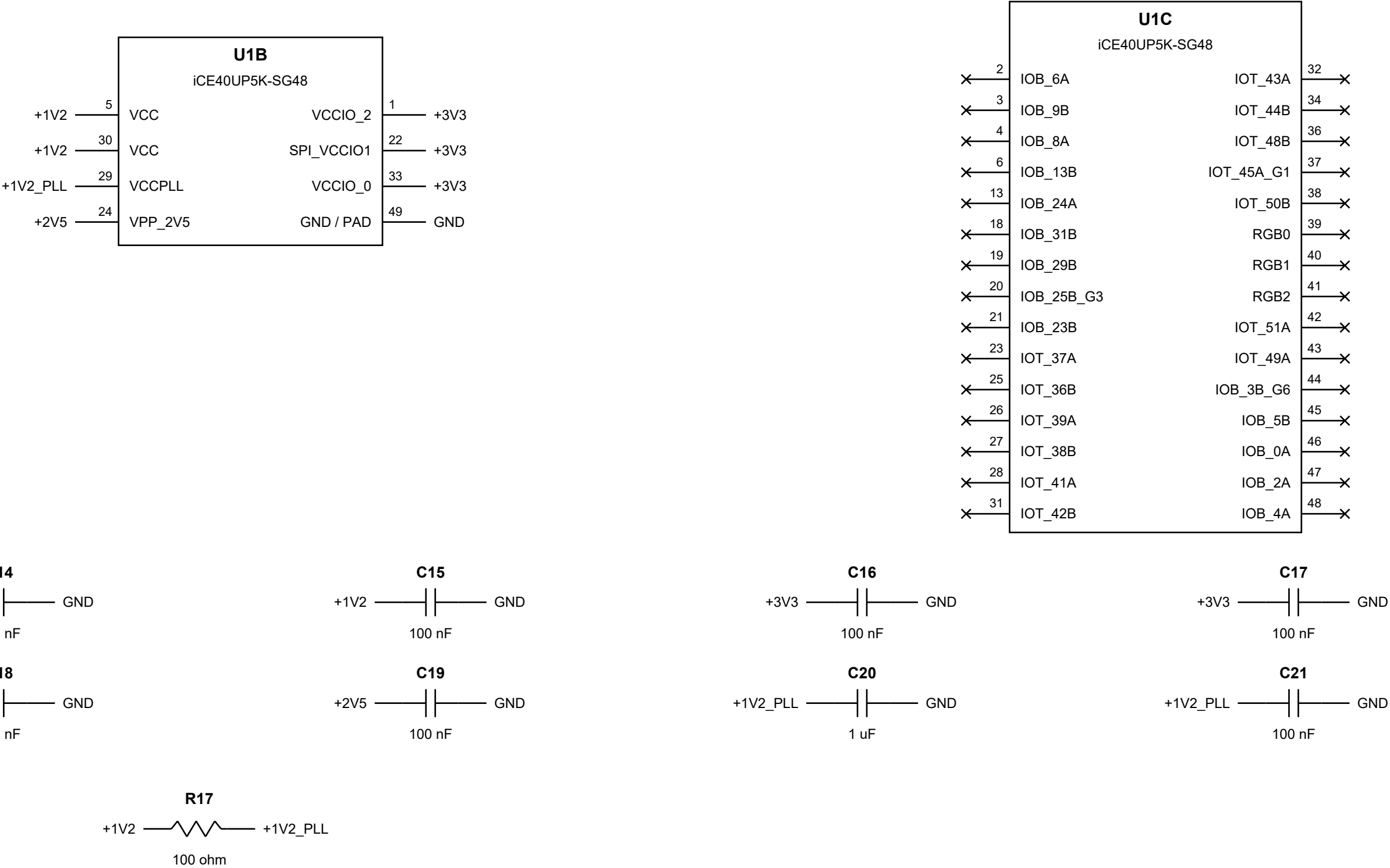
U5 releases IO_ENABLE after the 1.2 V core rail exceeds its threshold; U6 then enables 2.5 V. U7 holds CRESET_N low until 2.5 V is valid. CT = 1 nF gives about 6.2 ms nominal delay per supervisor. Pull-ups R10/R11 are on C03; R7 on C03 also pulls up U7 open-drain output. All NC pins stay open.

ENER / FPGA SERIAL LINK CONFIGURATION AND CLOCK

SPI0 is reserved for this point-to-point peripheral. Runtime mode 0, 1 MHz maximum.
Y1 drives the global-clock-capable pad 35 through R5. The internal PLL is unused.
CRESET_N is wired-OR: U7, SW1 and Pi GPIO17 can pull low. Pi drives low or releases to input; never drives high.

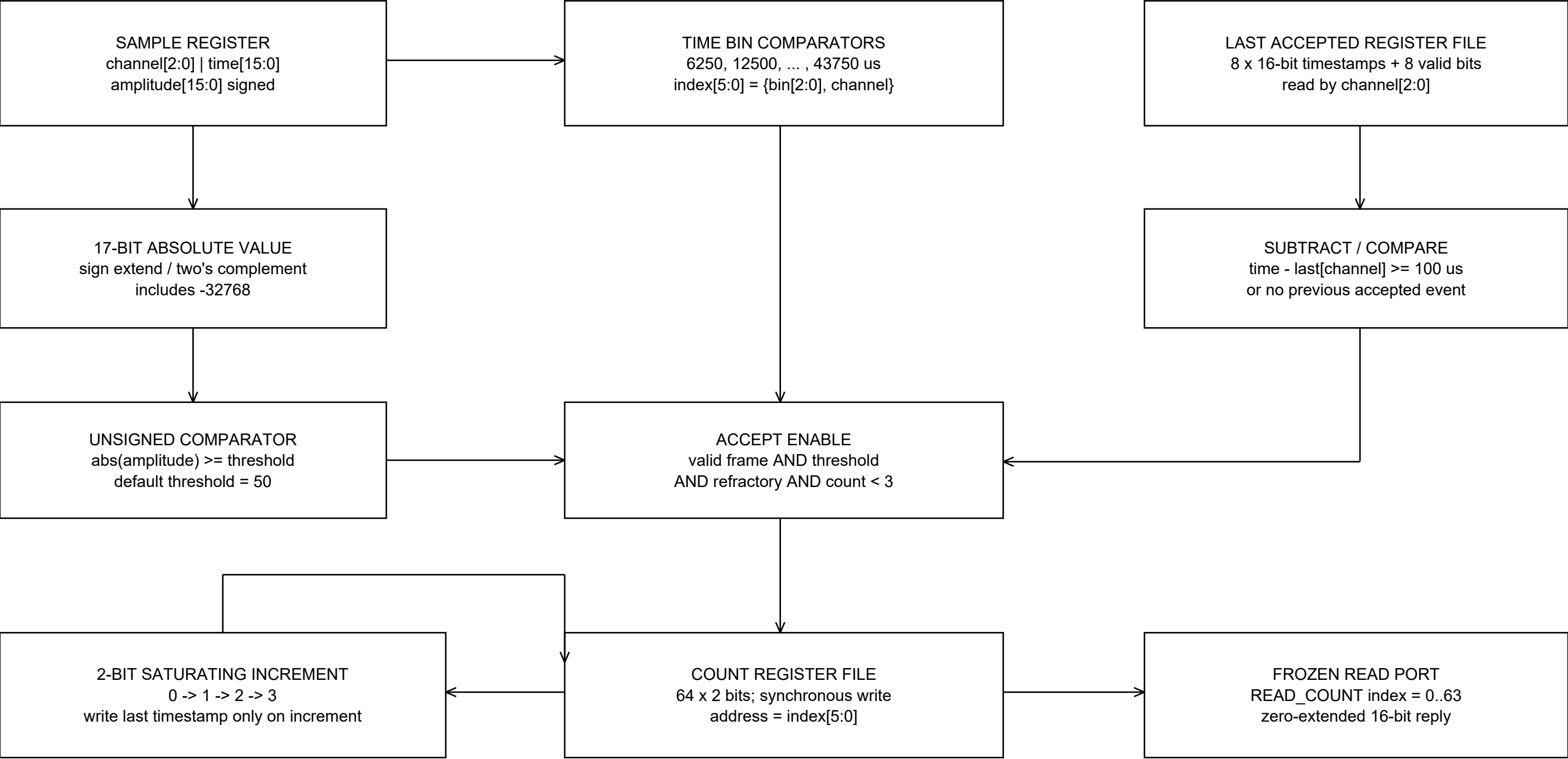


ENER / FPGA SUPPLY PINS BYPASS NETWORK AND UNUSED PADS



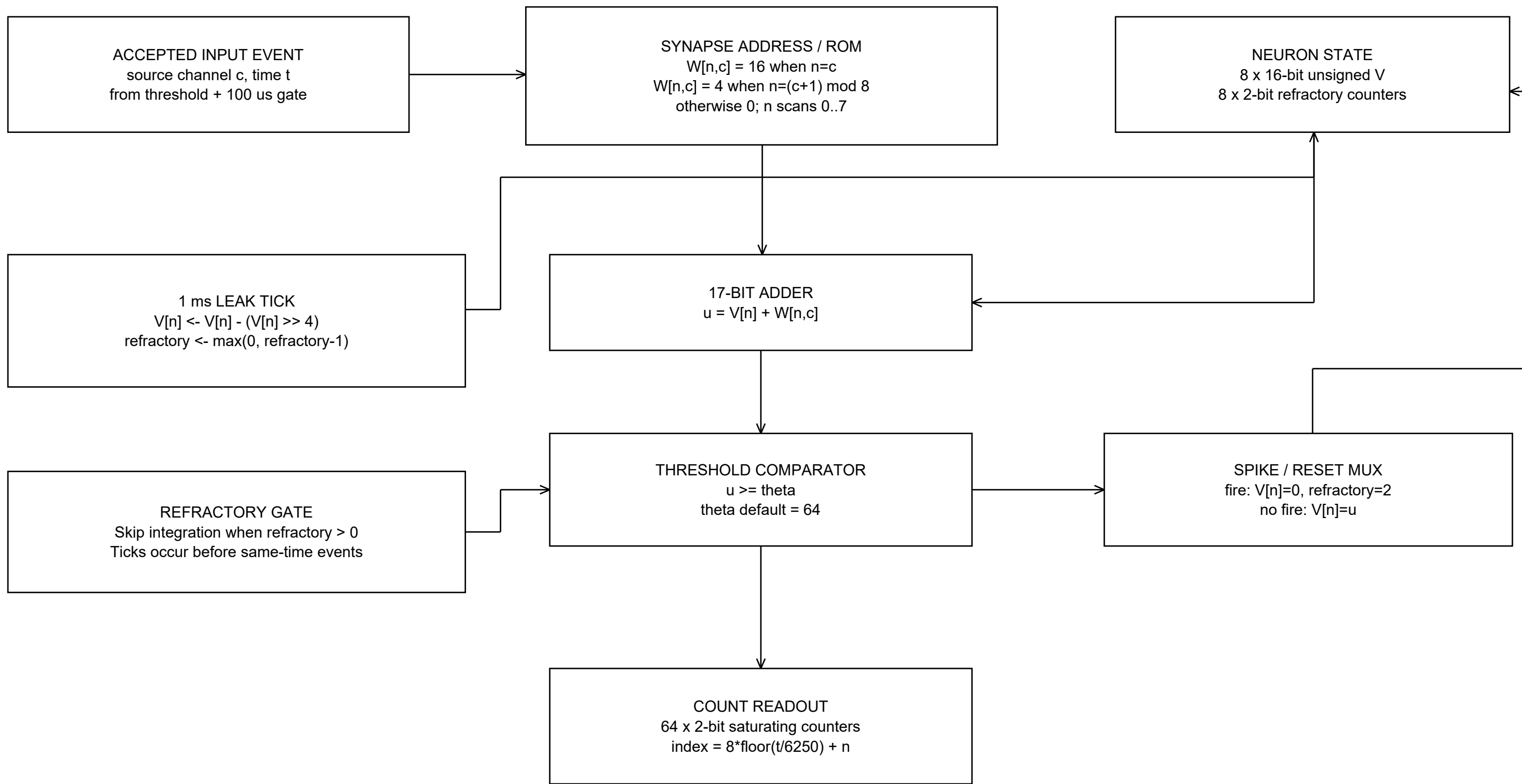
C14/C15 at core pins 5/30; C16/C17/C18 at I/O pins 1/22/33; C19 at VPP pin 24.
R17 with C20/C21 filters VCCPLL; connect the rail even though the PLL is unused.
Pad 49 is the exposed ground paddle: connect to the ground plane with a thermal/via array.
Unused pads shown with an X have no copper connection. Confirm unused-I/O constraints in the final bitstream.

ENER / SPATIAL EVENT DETECTION AND COUNTER CIRCUIT



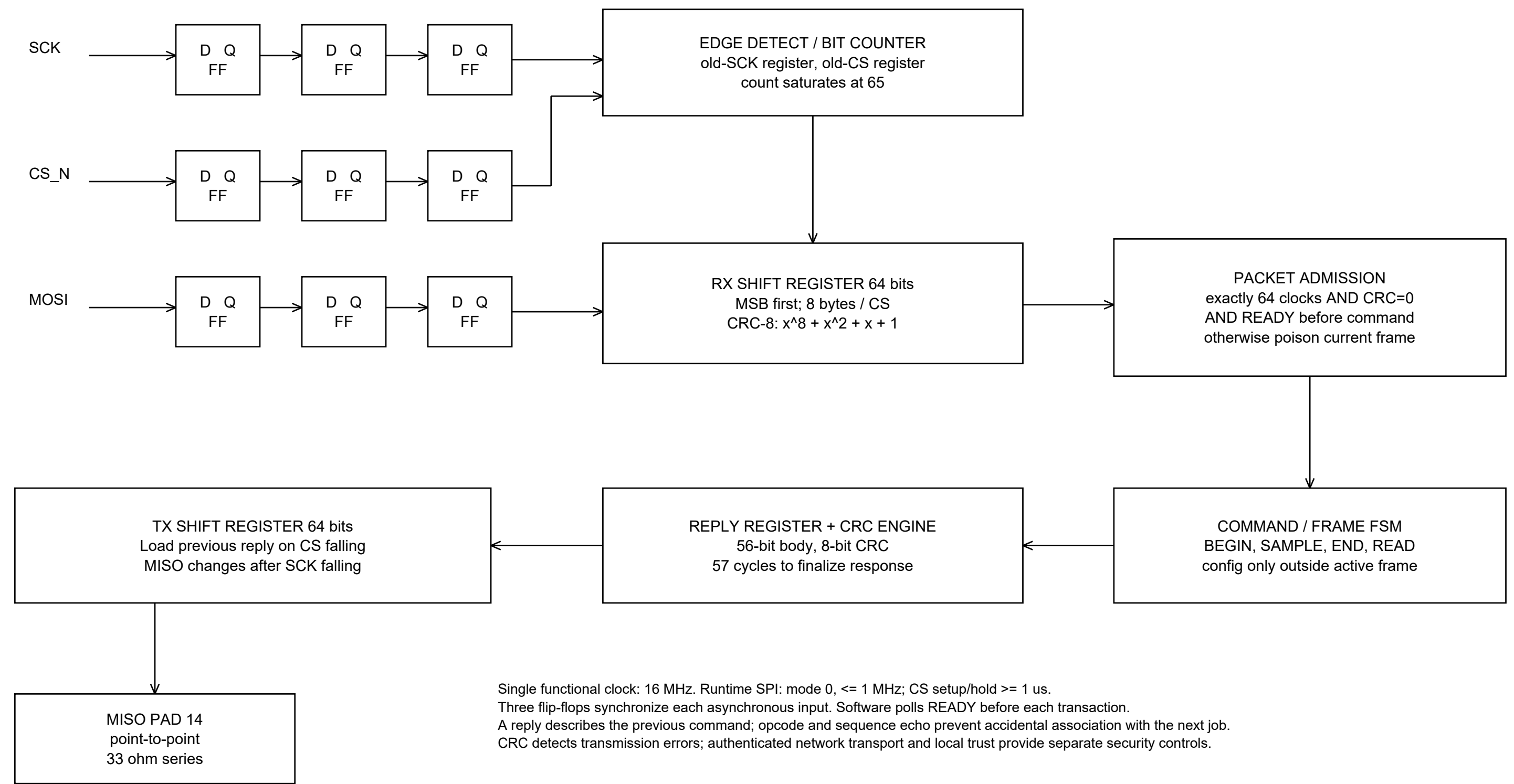
BEGIN clears all 64 counters and frame-local history; END freezes the store only if the entire frame is valid.
Input must be in nondecreasing timestamp order. Equal timestamps retain source order. The host performs sorting.
This is threshold-based spatial event detection. Channel identifiers do not imply isolated biological neurons.

ENER / EIGHT-NEURON INTEGRATE-AND-FIRE DATAPATH

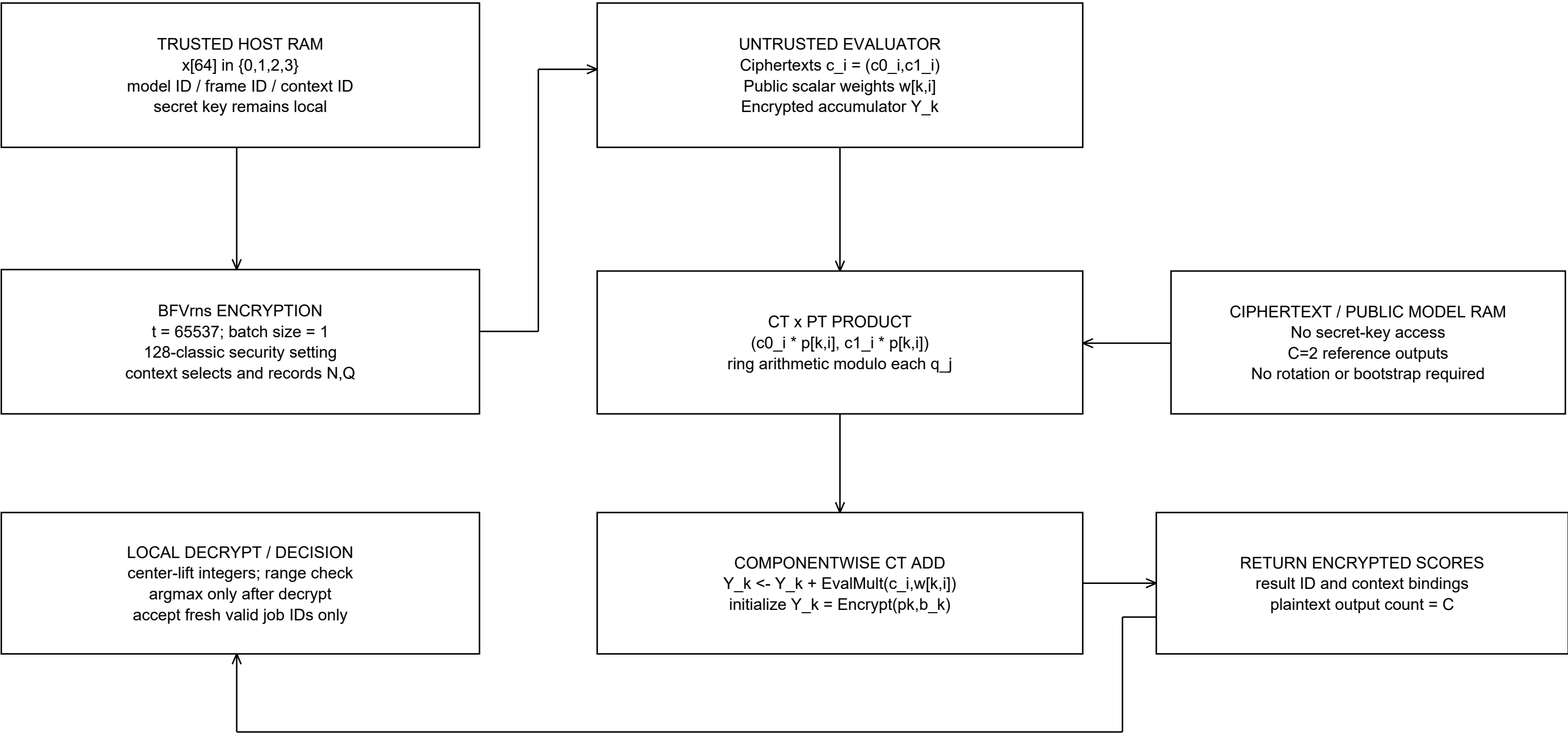


Every accepted source event visits eight output neurons. The two nonzero synapses form a local ring with self-input. The host sends timestamps relative to the frame; ADVANCE applies all missing millisecond ticks before integration. All neuron state resets on BEGIN. This explicit frame-local spiking profile requires its own classifier and calibration.

ENER / SERIAL TRANSPORT SHIFT REGISTERS AND CONTROL



ENER / FHE HOST MEMORY AND ENCRYPTED SCORE ARITHMETIC



The CPU executes this software arithmetic; U1 performs local neuromorphic encoding before encryption.
Exact sparse support is exposed only under an approved support-visible profile. The default exports all 64 encrypted positions.
Model bound: $\max_k (abs(b_k) + 3 \cdot \sum_i abs(w[k,i])) \leq 32768$. No modular-wrap ambiguity within this signed range.
The network protocol is a specified integration contract; the repository native demo executes client and evaluator in one process.